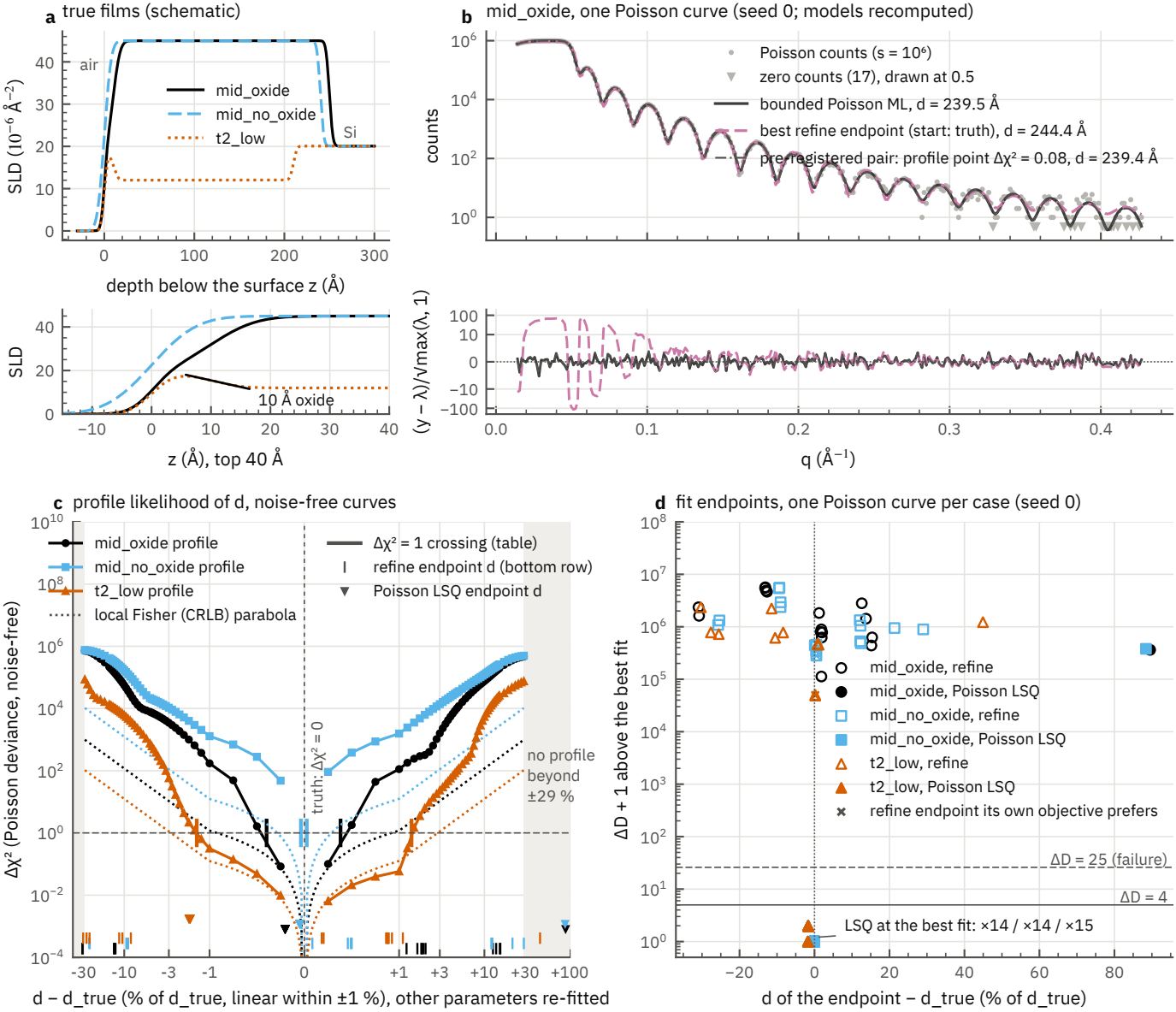

a true films (schematic)
air
mid_oxide
mid_no_oxide
t2_low
Si
SLD (10⁻⁶ Å⁻²)
depth below the surface z (Å)
SLD
10 Å oxide
z (Å), top 40 Å
b mid_oxide, one Poisson curve (seed 0; models recomputed)
Poisson counts (s = 10⁶)
zero counts (17), drawn at 0.5
bounded Poisson ML, d = 239.5 Å
best refine endpoint (start: truth), d = 244.4 Å
pre-registered pair: profile point Δχ² = 0.08, d = 239.4 Å
counts
(y − λ)/√max(λ, 1)
q (Å⁻¹)
c profile likelihood of d, noise-free curves
mid_oxide profile
mid_no_oxide profile
t2_low profile
local Fisher (CRLB) parabola
Δχ² = 1 crossing (table)
refine endpoint d (bottom row)
Poisson LSQ endpoint d
truth: Δχ² = 0
no profile beyond ±29 %
Δχ² (Poisson deviance, noise-free)
d − d_true (% of d_true, linear within ±1 %), other parameters re-fitted
d fit endpoints, one Poisson curve per case (seed 0)
mid_oxide, refine
mid_oxide, Poisson LSQ
mid_no_oxide, refine
mid_no_oxide, Poisson LSQ
t2_low, refine
t2_low, Poisson LSQ
refine endpoint its own objective prefers
ΔD = 25 (failure)
ΔD = 4
LSQ at the best fit: ×14 / ×14 / ×15
ΔD + 1 above the best fit
d of the endpoint − d_true (% of d_true)